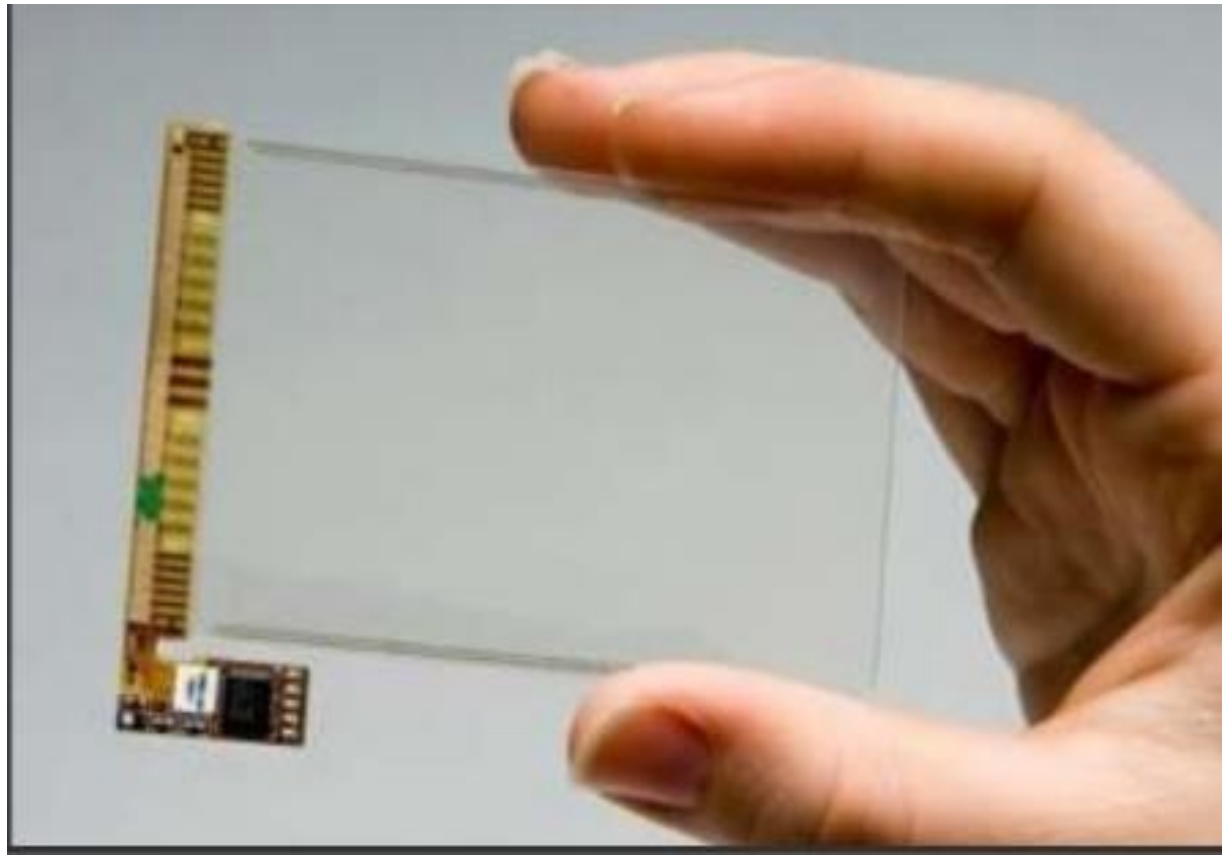
Touch Panel Control And Calibration

Touch Panel :

Touch Screen sensor is clear glass panel with a touch responsive surface, the sensor generally has an electrical current or signal going through it and touching the screen causes a voltage or signal change.



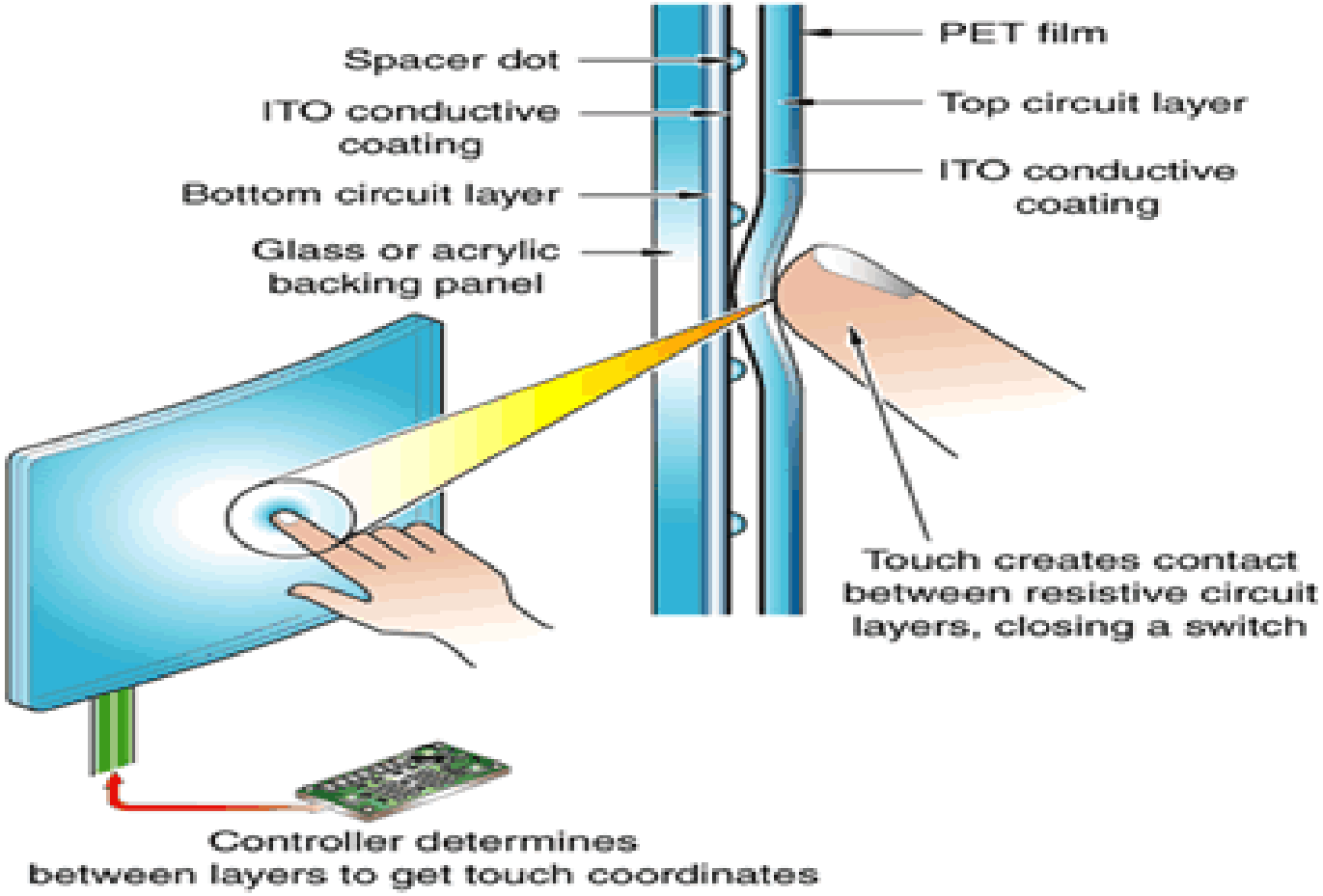
Types Of Touch Screen Technology

- **Resistive touch screen** :Resistive touch screen is composed of a flexible top layer made of polythene and a rigid bottom layer made of glass separated by insulating dots, attached to a touch screen controller.
- **Capacitive touch screen** : A capacitive touch screen panel is coated with a material that stores electrical charges.
- **Surface acoustic wave touch screen** : This technology contains two transducers placed along X-axis and Y-axis of the monitor's glass plate along with some reflectors. When the screen is touched, the waves are absorbed and a touch is detected at that point. These reflectors reflect all electrical signals sent from one transducer to another.
- **Infrared touch screen** : In infrared touch screen technology, an array of X and Y axis is fitted with pairs of IR LEDs and photo detectors. Photo detectors will detect any image in the pattern of light emitted by the LEDs whenever the user touches the screen.

FANUC Touch Panel

- FANUC touch Panel is **resistive type.**
- Touch panel can be used as an substitute to curser control for color LCD.
- The lower and side soft keys can be converted to touch panel operated soft keys.
- Touch Panel pressing information is read at intervals of **32 msec**.
- A positional precision of **+2.5mm** is provided.
- Touch panel function can be used with C-language executer.
- Touch panel is interfaced to System via **RS 232** communication port.
- When point is touched on touch Panel, it sends position co-ordinate to CNC via RS 232 communication.
- Co-ordinate System :The upper left corner of input area is set as origin of co-ordinate system .

Info-graphics For Resistive Type Touch Screen



Why Calibration Is Required ?

- Specifies an effective input area for touch.
- Specifies maximum co-ordinates.
- Corrects linearity.
- Makes panel input position match the LCD display position.
- Execution time improvement.

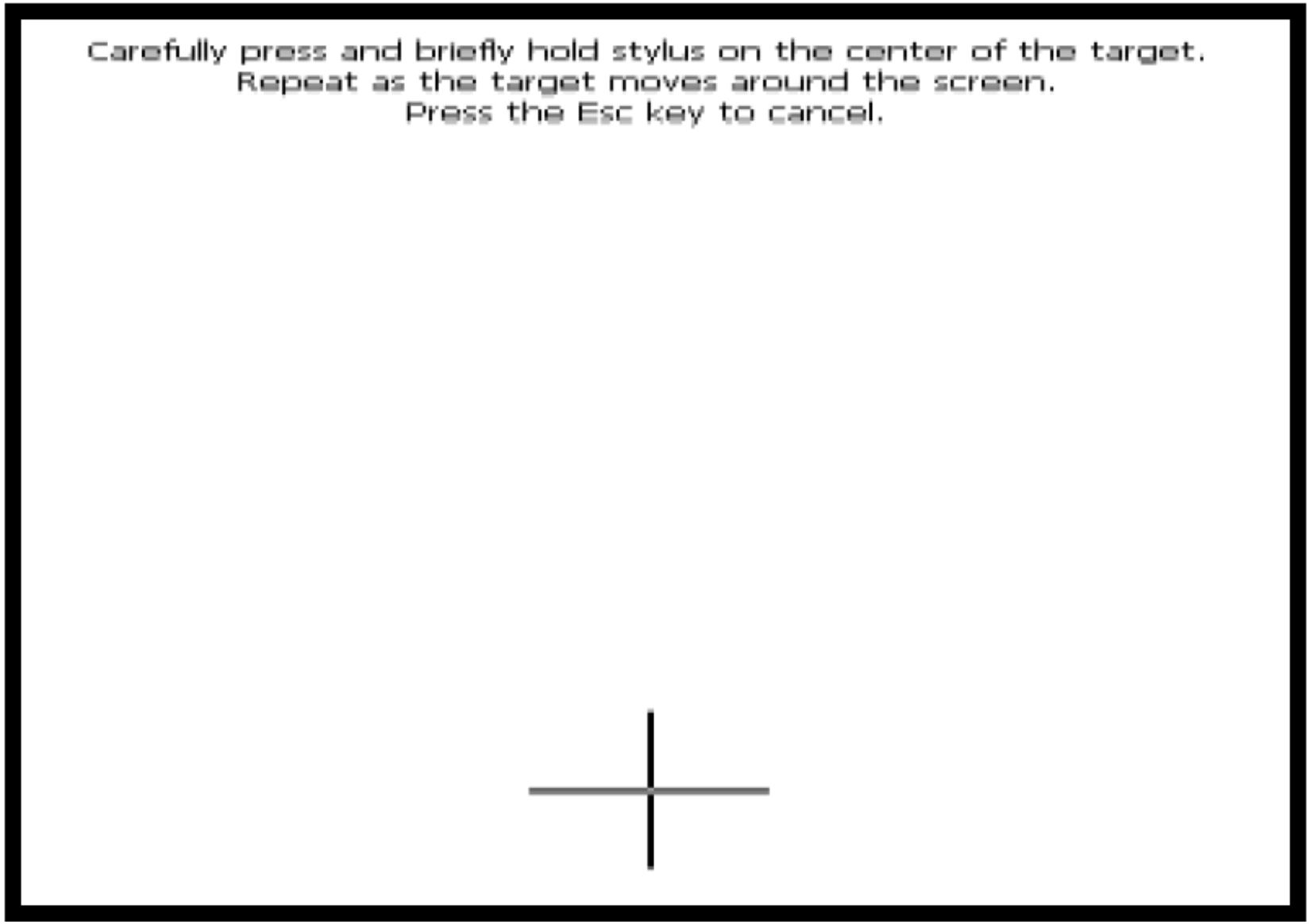
How To Calibrate Touch Panel

- Touch Panel calibration becomes necessary when touch panel is replaced or memory clear operation is performed.
- There are basically two types of configuration available in FANUC touch panel CNC.
    I. CNC with PC function
    II. Normal CNC
- Calibration can be performed in any mode.

For CNC With PC Function

- Press Alt + O to display following function menu.

- Select stylus to display screen as given below.

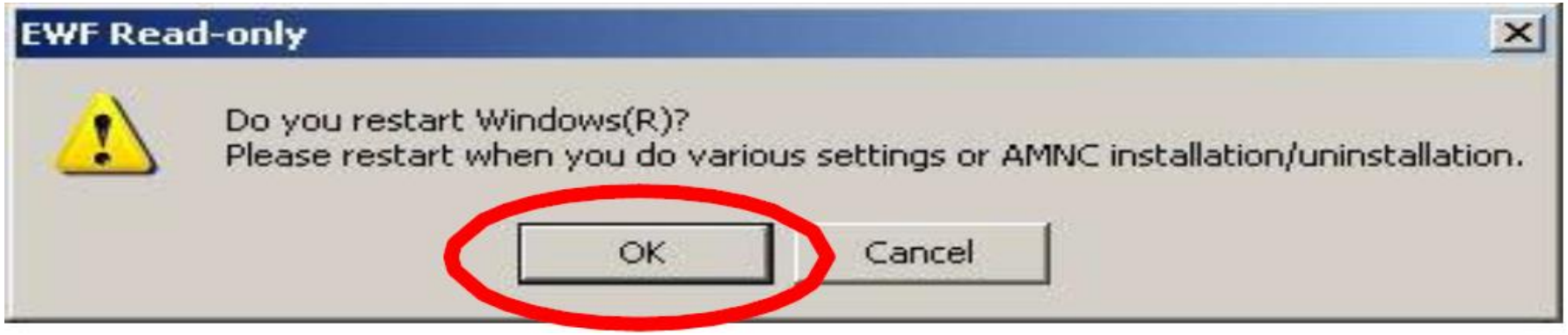

Carefully press and briefly hold stylus on the center of the target.
Repeat as the target moves around the screen.
Press the Esc key to cancel.

- Now when the + curser appears on screen, press the center point and hold till it disappears.
- Perform above step wherever curser appears and after completion, press return key.
- We can confirm the touch position after calibration completes.

In some of the Panel i, need to perform following procedure for calibration.

- Go to Compensation tool as follow:

- The message EWF read only will appear on screen, Press Ok.

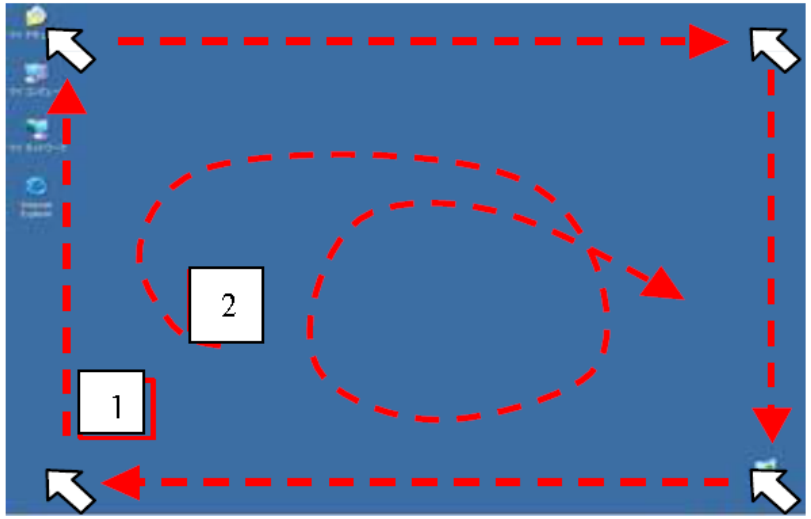


- Input User Name and Password and press ok.

- The windows OS will be activated as administrator, then open Compensation tool as mentioned in earlier step.
- Calibrate touch panel by touch red points wherever it appears.

- After completion of pressing all the cross hair points that appears on screen ,click on update which completes calibration process.

- We can confirm position shift of curser in each corner of screen and all area by moving cursor like 1 & 2.



Calibration Procedure For Normal CNC

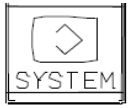
- For normal CNC , need to Enable touch panel calibration screen first, Set P3113#5(DCL) to 1

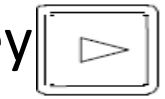
	#7	#6	#5	#4	#3	#2	#1	#0
3113	MS1	MS0	DCL					HMC

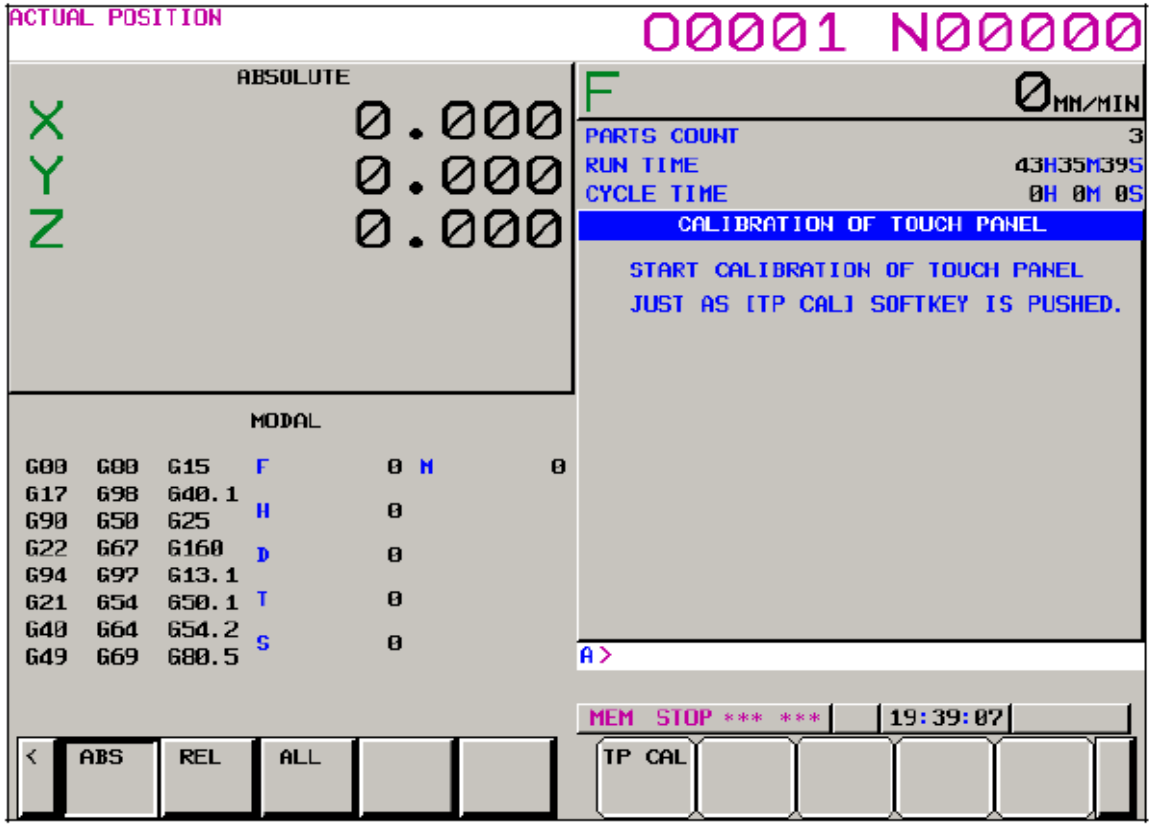
#5 **DCL** The touch panel compensation screen is:

0: Disabled.

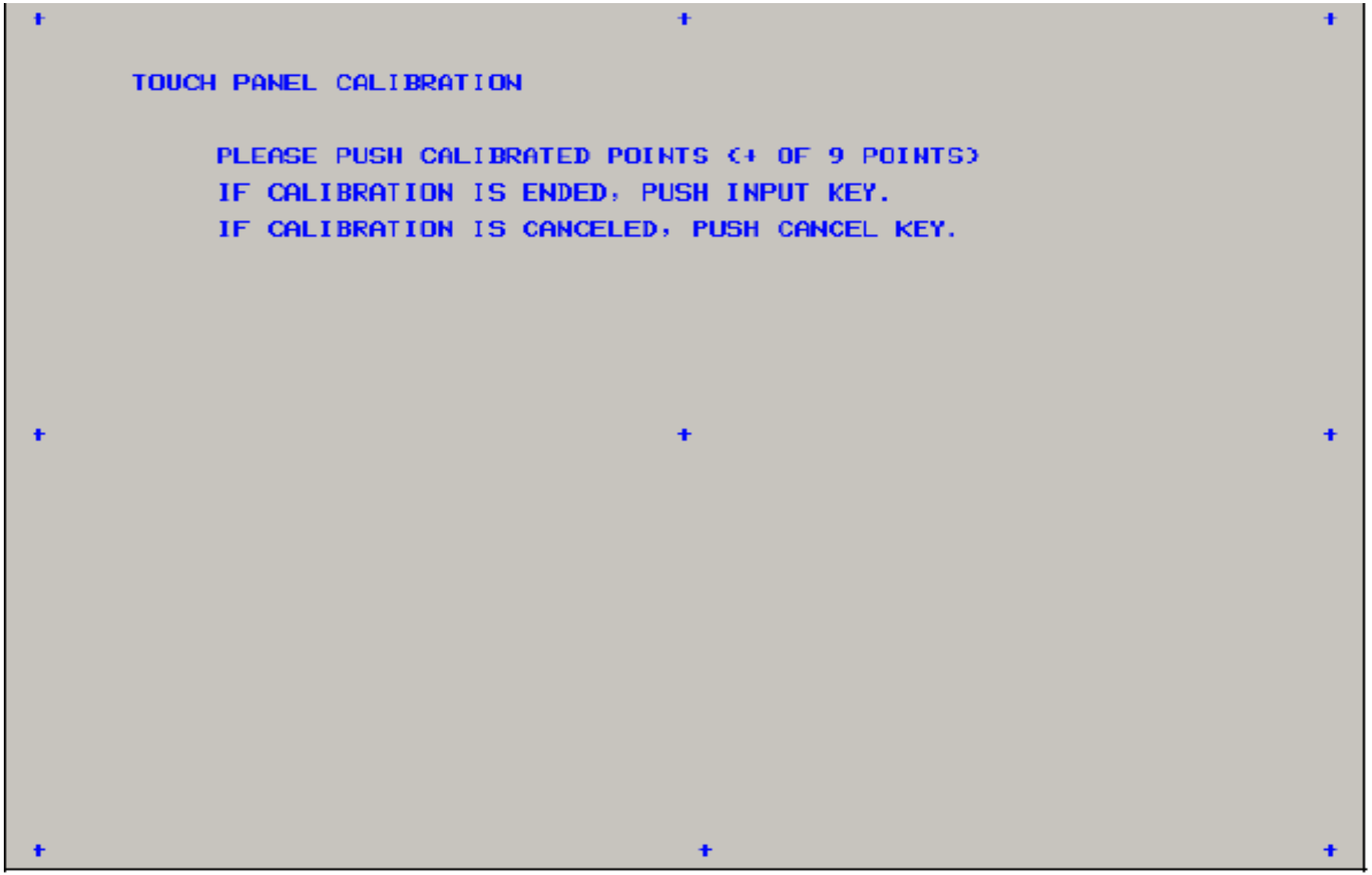
1: Enabled.

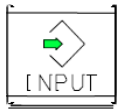
- Press function key 

- Press the continuous menu key  several times ,till touch panel soft key is displayed.
- Press [TOUCH PANEL] soft key and then press [OPRT]soft key.
- The [TP CAL] soft key is displayed.



- After pressing [TP CAL] soft key, it displays touch panel calibration screen.



- Press all the cross hair point displayed on screen with dedicated pen, when the point is pressed , the '+' mark changes to 'O' mark.
- If + mark is not pressed accurately, the message 'THE CALIBRATION POINT NOT MATCH, PLEASE PUSH AGAIN' is displayed.
- After pressing all calibration points, press input key 

to complete calibration.
- To cancel or retry calibration press cancel key

, the screen display returns to previous screen.
- If Input key is pressed before pressing calibration points, calibration operation is cancelled.
- When calibration terminated normally, the message 'CALIBRATION WAS ENDED' is displayed.
- Upon completion of calibration, disable touch panel calibration screen by P3113#5(DCL)=0.

Operation When Multiple Points Are Pressed.

- When two or more points are pressed ,then the position of gravity center is obtained by considering how each of these point pressed and touch panel is assume to be pressed at this gravity center position.
- Co-ordinates to be obtained after pressing multiple points at a time is decided by parameter P3192.

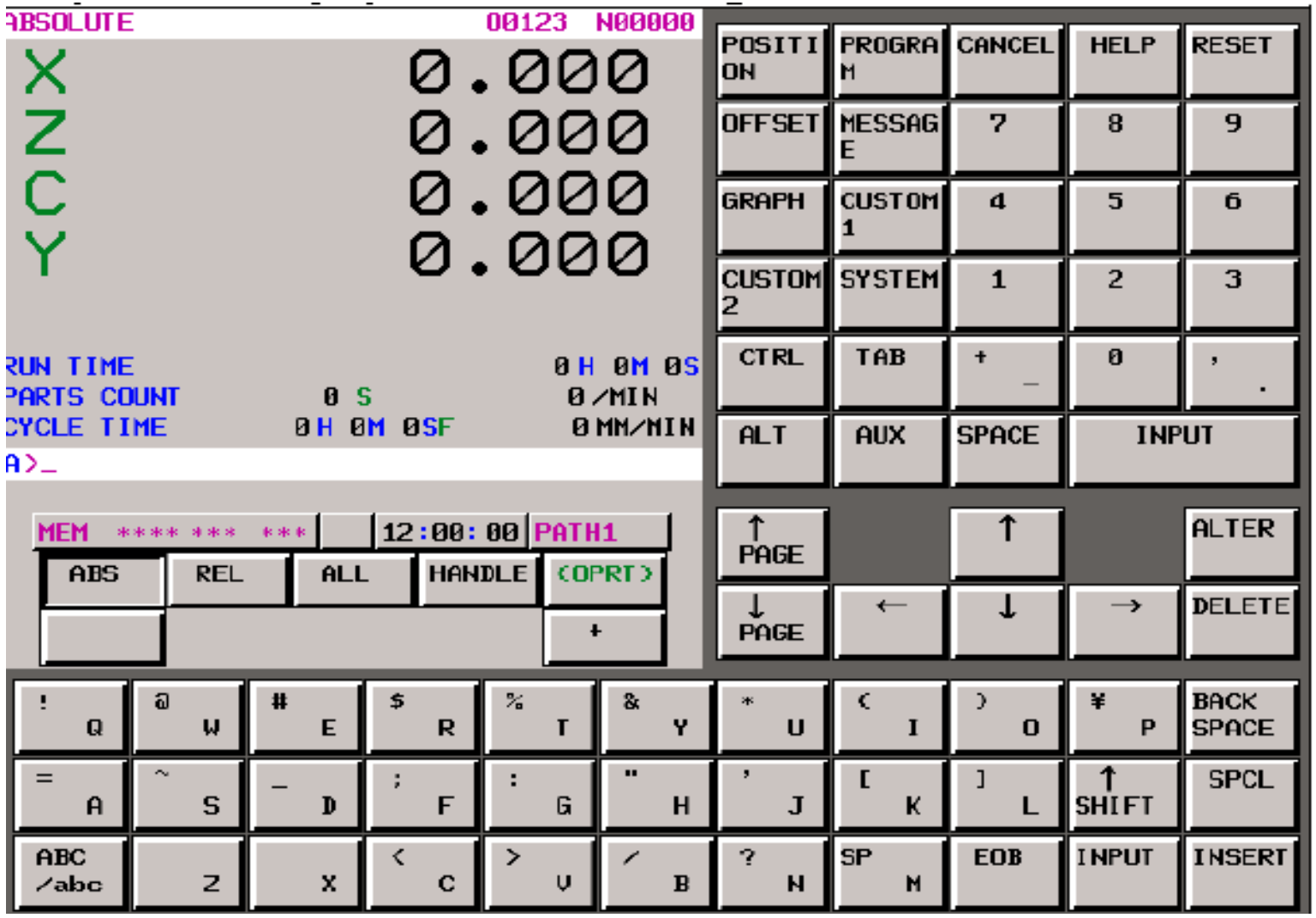
	#7	#6	#5	#4	#3	#2	#1	#0
3192						TRA	T2P	

[Input type] Parameter input
[Data type] Bit

#1 T2P When more than one point is pressed on the touch panel:
0: The position at the center of gravity is obtained.
1: The point pressed first is obtained.

NOTE
1 When two points are pressed in a row in a short time interval, the touch panel may react as if the center of the two points was pressed, even though parameter T2P (No. 3192#1) is set to 1.
2 If a C executer application or the like has a touch panel drag (move in pressed state) function, set parameter T2P to 0.

- In case of virtual MDI ,the operation to be performed for pressing two keys simultaneously such as ‘CAN’ and ‘RESET’ to clear alarm PS0100.



- Press SPCL key, the SPCL key held down and places system in SPCL mode.
- Press the key to be pressed simultaneously ,one at a time and press INPUT key.
- Of the key pressed in SPCL mode, The last two assumed to have been pressed simultaneously ,the system released from SPCL mode and returns to normal state.
- E.g. SPCL ➡ CAN ➡ RESET ➡ INPUT.

THANK YOU..!

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